ENGINEERING DUE DILIGENCE / UNITED STATES / 2026

Patent-Safety AI/ML Technique List

A patent-risk triage of 67 modern-ML and classical-AI techniques for an always-open-source commercial SaaS ecosystem.

Date stamped September 12, 2026

4 caution areas

identified in-force third-party patents or an actively issuing thicket

2 safest anchors

contributor grants plus confirmed-expired foundational patents

2 adjacent flags

continuous batching and asynchronous actor-critic require claim-scope review

67 evaluations

31 modern-ML techniques plus 36 classical-AI candidates across 10 branches

Engineering due diligence only — NOT legal advice, NOT freedom-to-operate clearance. Default jurisdiction: US. The patent landscape is perishable: patents issue and expire continuously; re-screen annually or before product launch.

The key correction. Every license patent grant here — Apache-2.0 §3, GPLv3 §11, and MPL-2.0 §2.1(b) — is contributor-scoped. It covers the contributors' patent claims, not third-party patents on the technique. The residual risk was always third-party patents; this edition itemizes it.

HOW TO USE THIS REFERENCE

The tiers separate license protection from technique-level risk

T1 — contributor patent grant. Confirmed in the implementation license. Valuable against contributors; silent about outside patentees.

T2 — foundational patents expired. Strong historical evidence that the original patent barrier has lapsed; improvements can still be patented.

T3 — credibly pledge-covered. A public promise or cross-license appears relevant. Scope, membership, and transfer conditions still matter.

T4 — published prior art; nothing asserted. The weakest positive tier. “No patents found” means medium confidence at best, not a clean bill of health.

T5 — caution. A known in-force third-party patent, aggressive holder, live assertion pattern, or dense issuing thicket warrants targeted review.

Modern-ML bottom line. Four areas land at T5: Coqui STT/Baidu, gensim/Google word2vec, Stable-Baselines3 DQN/Google, and latent diffusion's growing thicket. Two more carry grouped T5-adjacent flags: vLLM/SGLang through FriendliAI continuous batching, and RLlib async algorithms through DeepMind A3C. Essentia and Shogun are the two T1+T2 high-confidence anchors.

Classical-AI bottom line. Thirty-six candidates were assessed across 10 branches: nine T1 license grants, zero T2 assignments survived the refined bar, and three technique areas carry T5-adjacent carve-outs. Most remain T4, medium confidence at best. This validates the rule that an old foundation does not clear modern improvements.

What T1 really buys

It covers the party most likely to know its own contribution and patents. It is meaningful contractual protection, but not a technique-wide patent immunity.

What no tier buys

Claim construction, jurisdictional clearance, continuation-family review, or advice on a final product architecture. Those require counsel.

T5 CAUTION / IDENTIFIED THIRD-PARTY PATENTS

Speech recognition and word2vec carry the clearest claim-level exposure

Coqui STT — end-to-end speech recognition

T5 · HIGH CONFIDENCE

PAPER Deep Speech: Scaling up end-to-end speech recognition

REPOSITORY https://github.com/coqui-ai/STT

LICENSE MPL-2.0; canonical root `LICENSE`; express contributor grant in §2.1(b).

COMPATIBILITY MPL-covered files remain MPL and preserve Exhibit A notices; new files may remain Apache-2.0. SaaS adds no network-source duty.

PATENT EVIDENCE Baidu US 10,332,509 covers end-to-end recurrent-neural-network speech recognition; filed November 2016, granted June 2019, and in force to approximately November 2036. Baidu's December 2014 provisionals slightly predate the DeepSpeech paper. Coqui STT is in the same lineage. MPL's contributor grant cannot neutralize a third-party holder. This is the set's prime litigation-exposure item.

gensim — word2vec and topic modeling

T5 · HIGH ON PATENT EXISTENCE

PAPER Efficient Estimation of Word Representations in Vector Space

REPOSITORY https://github.com/RaRe-Technologies/gensim

LICENSE LGPL-2.1-only; root `COPYING`; no later-version grant and no express patent grant.

COMPATIBILITY Dynamic import into an Apache-2.0 application is the clean pattern if gensim remains LGPL and users can replace it; no SaaS network clause.

PATENT EVIDENCE Google US 9,037,464 covers skip-gram/CBOW word embeddings; filed March 2013, granted May 2015, and in force to approximately March 2033. gensim implements the claimed technique. Google has not asserted it. Mitigation: Google's own word2vec C implementation is Apache-2.0, so §3 covers Google's contributions and provides a shield gensim lacks.

> **Decision rule.** Do not treat public source or copyleft compliance as a patent defense. Before shipping either technique, obtain a focused claim-scope opinion or adopt the named lower-exposure implementation path.

T5 CAUTION / MIXED COMPONENTS AND PATENT THICKETS

Choose the lower-exposure algorithm, not merely the convenient library

Stable-Baselines3 — reinforcement-learning baselines

T5 · MEDIUM

PAPER Stable-Baselines3: Reliable Reinforcement Learning Implementations

REPOSITORY https://github.com/DLR-RM/stable-baselines3

LICENSE MIT; no express contributor patent grant.

COMPATIBILITY Easy to combine into Apache-2.0, but the permissive copyright license supplies no patent shield.

PATENT EVIDENCE Component split: DQN is T5 because Google/DeepMind US 9,679,258 B2 (priority October 2013, in force to about 2035; continuation US 11,049,008 B2) plausibly reads on the target-network mechanism. US 11,783,182 B2, in force to about 2037, sits near asynchronous A2C. There is no assertion history, but that is behavioral rather than legal protection. Prefer PPO, SAC, or TD3, which remain T4 with no patents found.

stable-diffusion-webui — latent diffusion

T5 · MEDIUM

PAPER High-Resolution Image Synthesis with Latent Diffusion Models

REPOSITORY https://github.com/AUTOMATIC1111/stable-diffusion-webui

LICENSE AGPL-3.0-only; root `LICENSE.txt`; GPLv3 patent grant applies.

COMPATIBILITY Provide corresponding source to network users. Keep the implementation outside an Apache-2.0 core unless the combined derivative is AGPL-compatible.

PATENT EVIDENCE No patent on latent diffusion models was found, and DDPM appears unpatented. The surrounding field is young and dense: Google US 12,136,141 B2, granted December 2024, covers text-to-image diffusion pipeline elements; Google and Adobe have pending applications citing DDPM/DDIM; Baidu and Tencent lead global diffusion-model patent families. Zero litigation so far is a weak signal while 2021–2024 filings are only now issuing.

> **Risk reduction.** Select by algorithm and deployment path. Swapping a library without changing the patented mechanism does not necessarily change the exposure.

T5-ADJACENT / INFERENCE SCHEDULING

Paged and radix caches look clear; continuous batching is the pressure point

vLLM / PagedAttention

T1 + CAUTION · MEDIUM-HIGH

PAPER Efficient Memory Management for Large Language Model Serving with PagedAttention

REPOSITORY https://github.com/vllm-project/vllm

LICENSE Apache-2.0; root `LICENSE` verified; express §3 contributor grant.

COMPATIBILITY Directly combinable with an Apache-2.0 core; preserve license and notices.

PATENT EVIDENCE No patents were found on PagedAttention or vLLM's block-table KV-cache scheme. FriendliAI's continuous-batching family sits at the scheduling layer: US 11,442,775 B2 (priority December 2021, granted September 2022, in force to about 2041), continuations US 11,836,520 B2 and US 12,524,266 B2, granted EP 4,191,474 B1, plus CN/KR counterparts. FriendliAI asserted the family against Hugging Face in D. Del. 1:23-cv-00816 in July 2023; the case settled after PTAB review.

SGLang / RadixAttention

T1 + CAUTION · MEDIUM-HIGH

PAPER SGLang: Efficient Execution of Structured Language Model Programs

REPOSITORY https://github.com/sgl-project/sglang

LICENSE Apache-2.0; root license verified; express §3 contributor grant.

COMPATIBILITY Directly combinable with an Apache-2.0 core; review separately licensed dependencies.

PATENT EVIDENCE No patents were found on radix-tree prefix caching or KV sharing. The same FriendliAI continuous-batching portfolio applies at the scheduling layer. Also track pending US 2025/0061316 A1, which addresses paged KV cache plus dynamic quantization; the assignee was not confirmed.

> **Launch gate.** Before commercial SaaS deployment, have counsel compare the actual scheduler against the FriendliAI claim families, including EU exposure and the newest continuations. The cache data structure and batching scheduler are separate review units.

T5-ADJACENT / TRAINING AND AGENTS

Algorithm selection and orchestration design change the exposure

Ray RLlib — distributed reinforcement learning

SPLIT: T1+T2 / T5 ASYNC

PAPER RLlib: Abstractions for Distributed Reinforcement Learning

REPOSITORY https://github.com/ray-project/ray

LICENSE Apache-2.0; repository license verified; express §3 contributor grant.

COMPATIBILITY Directly combinable with Apache-2.0. Audit the specific algorithms selected rather than treating RLlib as one patent unit.

PATENT EVIDENCE Classic Q-learning and policy-gradient families are T1+T2 with a strong 1990s-era expiry signal; DQN retains the separate caveat on page 4. Async actor-critic methods A3C, IMPALA, and APPO are T5: DeepMind/Google US 11,334,792 B2 (priority November 2015, in force to about 2036) and EP 3,360,085 B1 cover the family. The holder is not an aggressive assertor, which moderates but does not remove exposure.

smolagents — minimal agent framework

T1 + ADJACENT CAUTION

PAPER None; cite the project implementation.

REPOSITORY https://github.com/huggingface/smolagents

LICENSE Apache-2.0; root `LICENSE` verified; express §3 contributor grant.

COMPATIBILITY Directly combinable with an Apache-2.0 core.

PATENT EVIDENCE No CodeAct-specific patent was found; ReAct's October 2022 publication is useful prior art. C3.ai US 12,111,859, issued October 2024 and in force to about 2043, claims multi-agent orchestration with tool invocation. It is competitor-held and unasserted, but is the closest foundational agent-framework patent found. A claim read is prudent before commercializing multi-agent products.

T1 + T2 / STRONGEST HISTORICAL POSITION

Expired foundations strengthen the contributor-grant baseline

Essentia — audio and music-information retrieval

T1 + T2 · HIGH

PAPER *Essentia: An Audio Analysis Library for Music Information Retrieval* — Bogdanov et al., ISMIR 2013, pp. 493–498.

REPOSITORY https://github.com/MTG/essentia

LICENSE AGPL-3.0-or-later; `COPYING.txt` plus operative grant in `Essentia Licensing.txt`.

COMPATIBILITY Provide source to network users and keep the component separately licensed unless the combined derivative is AGPL-compatible. A commercial license is available.

PATENT EVIDENCE Historically important audio-fingerprinting patents are confirmed expired: Shazam US 6,990,453, filed in 2000 and expired around 2021, plus companions. The 2005-era Tune Hunter assertion patents are likewise expired; underlying audio features predate 2000.

Shogun — classical machine learning

T1 + T2 · HIGH

PAPER *The SHOGUN Machine Learning Toolbox* — Sonnenburg et al., JMLR 11(60):1799–1802, 2010.

REPOSITORY https://github.com/shogun-toolbox/shogun

LICENSE GPL-3.0-or-later; root `COPYING` is GPLv3 and file headers grant v3 or later.

COMPATIBILITY Keep separate from an Apache-2.0 core unless the combined work will be GPL-compatible. Audit optional `shogun-gpl` and per-component licenses.

PATENT EVIDENCE Foundational SVM patents are confirmed expired: US 5,649,068 (1993), US 5,950,146 (expired October 2016), and Burges-era filings around 2000 (expired around 2020). Health Discovery Corp v. Intel, the relevant assertion campaign, failed under §101/*Alice*.

Stanford CoreNLP — Java NLP pipeline

T1 + T2 · MEDIUM-HIGH

PAPER *The Stanford CoreNLP Natural Language Processing Toolkit* — Manning et al., ACL 2014, P14-5010.

REPOSITORY https://github.com/stanfordnlp/CoreNLP

LICENSE GPL-2.0-or-later by operative headers and README; the root file contains GPLv3 text.

COMPATIBILITY Elect GPLv3 to obtain §11's express patent grant; keep the component separate unless the combined work is GPL-compatible.

PATENT EVIDENCE No Stanford NLP patent surfaced. Tokenization, part-of-speech tagging, parsing, and named-entity recognition predate 2006, providing a strong expiry signal. The confidence is below the two confirmed-expiry entries because this is technique-age evidence rather than a complete family-by-family closure.

T1-BACKED / T4 LANDSCAPE

Kernels and memory optimizers have useful grants but incomplete third-party visibility

FlashInfer — attention kernels for serving

T1 · MEDIUM

PAPER FlashInfer: Efficient and Customizable Attention Engine for LLM Inference Serving

REPOSITORY https://github.com/flashinfer-ai/flashinfer

LICENSE Apache-2.0; root `LICENSE` verified. Bundled BSD-3-Clause and MIT components are identified in the license tail; preserve the NVIDIA/FlashInfer NOTICE.

COMPATIBILITY Directly combinable with Apache-2.0; audit `licenses/` and retain third-party attributions.

PATENT EVIDENCE The reported "FlashAttention patent filing" could not be verified: no published US application by Tri Dao or Stanford was found. Risk is structurally moderated by online-softmax prior art (Milakov and Gimelshein, 2018), longstanding tiling methods, universal adoption, and zero assertions. A filing less than about 18 months old could still be unpublished.

DeepSpeed / ZeRO — distributed training memory

T1 + T3/T4 · MEDIUM

PAPER ZeRO: Memory Optimizations Toward Training Trillion Parameter Models

REPOSITORY https://github.com/deepspeedai/DeepSpeed

LICENSE Apache-2.0; root `LICENSE` verified; express §3 grant.

COMPATIBILITY Directly combinable with Apache-2.0. Review ZeRO-Offload and ZeRO-Infinity revisions and dependencies separately.

PATENT EVIDENCE No ZeRO-specific patent was found, with medium confidence because Microsoft files extensively. Microsoft joined Open Invention Network (OIN) in October 2018 and LOT Network, but those arrangements are structural shields, not technique-specific clearance.

LoRA via Hugging Face PEFT

T1 + T4 · MEDIUM

PAPER LoRA: Low-Rank Adaptation of Large Language Models

REPOSITORY https://github.com/huggingface/peft

LICENSE Apache-2.0; root `LICENSE` verified; express §3 grant.

COMPATIBILITY Directly combinable with Apache-2.0. Evaluate other PEFT methods independently.

PATENT EVIDENCE No patent was found on base LoRA. Watch NVIDIA's pending US 2025/0252301 A1 on DoRA, weight-decomposed low-rank adaptation. DoRA is a distinct variant, not base LoRA; track prosecution if it enters the product.

T1-BACKED / T4 LANDSCAPE

Post-training frameworks are licensed clearly; their newest algorithms remain hard to screen

Hugging Face TRL — PPO, DPO, GRPO, and KTO

T1 + T4 · MEDIUM TO MEDIUM-LOW

PAPER No single canonical framework paper; TRL documentation links each algorithm to its original paper.

REPOSITORY https://github.com/huggingface/trl

LICENSE Apache-2.0; root `LICENSE` verified; express §3 grant.

COMPATIBILITY Directly combinable with Apache-2.0. Review each algorithm as its own patent unit.

PATENT EVIDENCE No patents were found on PPO, DPO, GRPO, or KTO. Confidence is medium for PPO and medium-low for the 2023–2024 DPO/GRPO/KTO methods because recent applications can remain unpublished. OpenAI's "defensive" patent pledge is a website statement, not a contract; do not count it as T3.

OpenRLHF — scalable RLHF infrastructure

T1 + T4 · MEDIUM

PAPER OpenRLHF: An Easy-to-use, Scalable and High-performance RLHF Framework

REPOSITORY https://github.com/OpenRLHF/OpenRLHF

LICENSE Apache-2.0; root `LICENSE` verified; express §3 grant.

COMPATIBILITY Directly combinable with Apache-2.0, but its Ray, vLLM, and DeepSpeed layers retain their own patent and dependency reviews.

PATENT EVIDENCE Its algorithm-level questions substantially overlap TRL. No OpenRLHF-specific patent or litigation was found. Composition does not merge the separate risk conclusions for the scheduler, training engine, or selected optimization algorithm.

> **Practical review unit.** Treat a framework as a bill of materials, not one technique. Record the trainer, optimizer, preference algorithm, distributed runtime, and serving scheduler separately.

T1-BACKED / AGENT FRAMEWORKS

Contributor grants are strong; third-party orchestration claims remain the common dependency

Aider — repository-aware coding loops

T1 · HIGH ON GRANT

PAPER None; cite the project implementation.

REPOSITORY https://github.com/paul-gauthier/aider

LICENSE Apache-2.0; root `LICENSE.txt` verified; express §3 grant.

COMPATIBILITY Directly combinable with Apache-2.0.

PATENT EVIDENCE No patent was found on repo maps, edit formats, or the agentic edit-lint-test loop. Adjacent Microsoft and Amazon code-generation filings are applications rather than grants and are model-centric rather than loop-centric.

Google Agent Development Kit

T1 · HIGH ON GRANT

PAPER None; cite the project implementation.

REPOSITORY https://github.com/google/adk-python

LICENSE Apache-2.0; root `LICENSE` verified; express §3 grant.

COMPATIBILITY Directly combinable with Apache-2.0.

PATENT EVIDENCE No ADK-specific patent or suit was found. Google's 2013 Open Patent Non-Assertion Pledge does not help here: its named list covers 10 MapReduce patents and 79 IBM/CA data-center patents, with no AI coverage.

Strands Agents SDK

T1 · HIGH ON GRANT

PAPER None; cite the project implementation.

REPOSITORY https://github.com/strands-agents/harness-sdk

LICENSE Apache-2.0 in `LICENSE.APACHE`; preserve the Amazon NOTICE.

COMPATIBILITY Directly combinable with Apache-2.0; preserve the NOTICE on redistribution.

PATENT EVIDENCE No Strands-specific patent or suit was found. Residual exposure is the generic orchestration category, including C3.ai US 12,111,859 discussed with smolagents.

T1-BACKED / RETRIEVAL AND EVALUATION

Evaluation carries little claim surface; RAG deserves a live litigation watch

Haystack — retrieval-augmented generation

T1 · WATCH-LIST

PAPER None; cite the project implementation.

REPOSITORY https://github.com/deepset-ai/haystack

LICENSE Apache-2.0; root `LICENSE` verified; express §3 grant.

COMPATIBILITY Directly combinable with Apache-2.0.

PATENT EVIDENCE Generic RAG was published by Meta in 2020 with no known filing, but this is the most actively patented area in the group. Cohesity US 12,619,501, issued May 2026 and in force to about 2046, concerns RAG over backup data. Many Worlds v. OpenAI, filed in the Eastern District of Texas around September 2, 2026, asserts five patents with March 2011 priority on retrieval, personalization, and generation. Watch the docket.

Terminal-Bench — terminal-agent evaluation

T1 · HIGH

PAPER Terminal-Bench: Benchmarking Agents on Hard, Realistic Tasks in Command Line Interfaces

REPOSITORY https://github.com/laude-institute/terminal-bench

LICENSE Apache-2.0; root `LICENSE` verified; express §3 grant.

COMPATIBILITY Directly combinable with Apache-2.0.

PATENT EVIDENCE An evaluation harness has little patent claim surface. No technique-specific patent or litigation was found. Benchmark contents can still raise copyright and data-license questions, which are outside this patent tier.

EnvACE — world rehearsal for agentic RL

T1 · HIGH ON GRANT

PAPER EnvACE: Internalizing Environment Dynamics via World Rehearsal for Agentic Reinforcement Learning

REPOSITORY https://github.com/Within-yao/EnvACE

LICENSE Apache-2.0; root `LICENSE` verified; express §3 grant.

COMPATIBILITY Directly combinable with Apache-2.0.

PATENT EVIDENCE No patent was found. The paper is roughly five weeks old as of this edition, creating the full unpublished-application blind spot; “high” refers to the license grant, not technique clearance.

T1-BACKED / VISION AND SPEECH

Known prior art helps YOLO; newer voice systems remain surrounded by active filings

Ultralytics / YOLO — real-time object detection

T1 + T4

PAPER You Only Look Once: Unified, Real-Time Object Detection

REPOSITORY https://github.com/ultralytics/ultralytics

LICENSE AGPL-3.0-only; root `LICENSE`; commercial license available.

COMPATIBILITY For SaaS, offer corresponding source to network users. Keep the component separate unless the combined derivative can be AGPL-compatible, or obtain a commercial license.

PATENT EVIDENCE No patent was found on YOLO itself or held by Ultralytics; public activity is trademark-related. YOLO is repeatedly cited as known prior art in later patents. The improvement-patent field is dense, but no identified claim targets the core method. T1 confidence is high; T4 confidence is medium.

Coqui TTS — zero-shot synthesis and conversion

T1 + T4 · MEDIUM

PAPER YourTTS: Towards Zero-Shot Multi-Speaker TTS and Zero-Shot Voice Conversion for everyone

REPOSITORY https://github.com/coqui-ai/TTS

LICENSE MPL-2.0; canonical root `LICENSE.txt`; express §2.1(b) grant.

COMPATIBILITY MPL-covered files stay MPL and retain notices; independently authored files may remain Apache-2.0. No network clause.

PATENT EVIDENCE No YourTTS-specific patent was found. The zero-shot voice field has 2023 applications and neighboring non-practicing-entity litigation, including SoundClear v. Google/Amazon and Parus Holdings v. Google.

Piper / VITS — end-to-end text to speech

T1 + T4 · MEDIUM

PAPER Conditional Variational Autoencoder with Adversarial Learning for End-to-End Text-to-Speech

REPOSITORY https://github.com/OHF-Voice/piper1-gpl

LICENSE GPL-3.0-only, read conservatively from root `COPYING`; the archived `rhasspy/piper` lineage is MIT.

COMPATIBILITY Keep the active fork separately licensed unless the combined work will be GPL-compatible. Ordinary SaaS use does not trigger GPL source distribution.

PATENT EVIDENCE No VITS-specific patent was found. The background neural-TTS field is crowded with big-technology patents, including US 12,020,685.

> **Model terms remain separate.** These conclusions cover implementation code and technique patents. Model weights, training voices, datasets, publicity rights, and trademarks require their own review.

APPENDIX / MIT IMPLEMENTATIONS

Permissive copyright terms do not include an express patent grant

All seven entries are MIT-licensed and can be combined into an Apache-2.0 codebase with attribution. None starts with T1. Their tiers describe the technique landscape, not a license shield.

Project and source	Tier	Patent note
llama.cpp GGUF inference	T4 · medium-low	No technique-specific patent found; dense quantization background, including Qualcomm US 10,373,050 B2 to about 2035.
OpenHands paper	T5 · structural	Clean technique landscape, but MIT supplies no shield against present or future third-party autonomous-agent patents.
SWE-bench	T4 · medium	No methodology patent found; benchmark methods are among the least patent-prone categories. Contents can raise copyright issues.
Stable-Baselines3	T5 · medium	DQN faces US 9,679,258 B2 to about 2035. Prefer PPO, SAC, or TD3, each T4 with no patent found.
Activity Frames paper / implementation reference	T4 · medium	No patent found; the approximately five-week-old paper has a full unpublished-application blind spot.
tau-bench / tau2-bench	T4 · medium	No methodology patent found; only a distant enterprise chatbot-scoring thicket.
ExLlamaV2	T4 · medium-low	No GPTQ patent found; dense quantization background. Root MIT license verified after an earlier GPL assumption was corrected.

Reimplementation tradeoff. Rewriting MIT, GPL, or AGPL code independently can avoid copyright-license inheritance, but it does not confer the original project's patent grant and does not avoid third-party claims.

ARCHITECTURE AND REIMPLEMENTATION

Open source makes disclosure manageable; it does not make patent risk disappear

Directly combinable

Apache-2.0: permissive with an express contributor grant. **MPL-2.0:** keep covered files and notices under MPL; new files may remain Apache-2.0. **LGPL:** dynamic linking is usually the clean boundary, subject to replacement and modification rights.

Keep separately licensed

GPLv3: a combined derivative must use GPL-compatible terms. **AGPLv3:** the same whole-work boundary plus a network source offer. A process or service boundary is conservative architecture, not an automatic legal safe harbor.

SaaS nuance. GPLv3 generally has no network-use trigger; AGPLv3 does. An always-public project can satisfy the practical source-availability duty with a compliant source link, but that does not let an AGPL combined work remain Apache-2.0.

Clean-room reimplementation preserves independence, not patent rights

1. Copyright protects the code's expression, not the underlying idea, algorithm, method, or fact. A genuinely independent implementation need not inherit the source code's license.
2. "Using code as a rough guide" is risky when structure, sequencing, naming, or other protectable expression carries over. A stronger process has one team produce a functional specification from papers and observable behavior and another implement without consulting the original code.
3. If the implementation is derivative enough to rely on the original license, its patent grant and its copyleft obligations arrive together. The grant cannot be kept while the duties are discarded.
4. If the implementation is truly independent, the original license does not attach — and neither does its contributor patent grant. Reimplementation cannot defeat a third-party patent on the technique.

Counsel point. Derivative-work status, linkage boundaries, network-source compliance, and patent claim scope are fact-specific. Document provenance and design decisions while the evidence is fresh.

METHOD AND LIMITS

A verified license file is only the first layer of diligence

Repository review

- Record the exact commit or release.
- Read the root license, operative file headers, README grant language, NOTICE files, and bundled-component licenses.
- Check model weights, datasets, trademarks, and dependencies separately.
- Repeat after material upgrades or architectural changes.

Patent review

- Search issued patents, applications, continuations, assignments, litigation, and relevant non-US families.
- Map claims to the product's actual mechanism, not merely to a paper title.
- Treat an approximately 18-month publication delay as a blind spot.
- Repeat annually and before launch.

Search boundary. This edition uses web and Google-Patents-indexed research, not a professional clearance search. Non-English filings, unpublished applications, continuations, assignment changes, and claim-construction issues may be missing. "No patent found" is medium confidence at best.

CoreNLP shows why boilerplate is not the operative grant

Stanford CoreNLP's root `LICENSE.txt` contains GPLv3 text, while its file headers and README grant GPL-2.0-or-later. Electing v3 supplies the §11 patent grant; relying only on the filename or boilerplate could produce the wrong conclusion. Review the grant notice that actually applies to the code.

Confidence labels

High is reserved for verified license grants, identified live patents, or confirmed expirations. **Medium** reflects a meaningful indexed search with no complete claim-level clearance. **Medium-low** marks a young or patent-dense field with larger blind spots.

PART III / CLASSICAL AI BRANCHES

Old foundations do not imply a clear modern patent field

Classical AI remains directly useful for exact optimization, formal verification, declarative reasoning, auditable rules, uncertainty, planning, and structured neural systems.

Refined method. Foundational-patent expiry was not treated as field clearance. Every branch was screened for post-2010 improvement patents, continuation and divisional activity, the "YOLO pattern" of an old core surrounded by live improvement claims, and non-practicing-entity assertion activity. T2 required entry-specific clearance evidence; ties resolved to T4.

10 branches

from SAT/SMT and proof assistants through neuro-symbolic integration

9 T1 grants

license-level contributor protection among the strongest candidates

0 T2 survivors

no classical entry met the refined field-clearance bar

8 of 10

branches showed the YOLO pattern: old core, live improvement thicket

What T4 means here

The core technique is well-published and no direct assertion was found, but modern implementations or applications may sit near active claims. Confidence is medium at best.

What T1 means here

The implementation license carries an express contributor grant. It improves the contract position without clearing unrelated patentees or later application-layer claims.

Positive pattern. Lean 4, Soufflé, and OR-Tools pair strong open licenses with favorable searches. They are engineering starting points, not freedom-to-operate opinions.

Exact solving is low-risk at the core; SMT configuration retains an active fringe

MiniSAT — embeddable CDCL SAT solver

T4 · MEDIUM

PAPER Eén and Sörensson, *An Extensible SAT-solver*, SAT 2003.

REPOSITORY https://github.com/niklasso/minisat

LICENSE MIT; verified; no express patent grant.

COMPATIBILITY Embeds directly into an Apache-2.0 codebase.

PATENT EVIDENCE The closest CDCL-era patent, US 7,401,305 B2 (IBM to Google), expired February 20, 2026. Qualcomm US 11,481,468 B2, active to December 2037, addresses distributed solver ensembles rather than single-threaded CDCL. No litigation was found.

CaDiCaL — incremental CDCL and proof logging

T4 · MEDIUM

PAPER Biere et al., *CaDiCaL 2.0*, CAV 2024.

REPOSITORY https://github.com/arminbiere/cadical

LICENSE MIT; verified; no express patent grant.

COMPATIBILITY Embeds directly; IPASIR and LRAT/DRAT support make it useful for proof-producing optimization.

PATENT EVIDENCE No patent or assertion was found against CaDiCaL or Kissat. The modern implementation remains T4 because current solver heuristics and proof systems continue to evolve.

Z3 — industrial SMT solver

T4 · MEDIUM

PAPER de Moura and Bjørner, *Z3: an efficient SMT solver*, TACAS 2008.

REPOSITORY https://github.com/Z3Prover/z3

LICENSE MIT © Microsoft; verified; no express patent grant.

COMPATIBILITY Embeds directly into Apache-2.0.

PATENT EVIDENCE No Z3 litigation was found. Microsoft US 9,489,221 B2 on matching-based SMT pattern inference lapsed, but adjacent patents US 8,387,021 and US 8,180,786 B2 and DNN-based SMT configuration applications remain. This is the branch's highest residual risk.

Proof assistants can turn transformations into checkable artifacts

Lean 4 — dependently typed language and prover

T1 · HIGH

PAPER de Moura and Ullrich, CADE 28, 2021.

REPOSITORY https://github.com/leanprover/lean4

LICENSE Apache-2.0; verified; express §3 grant.

COMPATIBILITY Merges directly into an Apache-2.0 codebase.

PATENT EVIDENCE No post-2010 prover-mechanism patent, thicket, or NPE activity was found. This is the strongest license-and-landscape posture among all entries in this edition.

E — saturation-based first-order prover

T4 · MEDIUM

PAPER Schulz, *AI Communications* 15(2/3), 2002.

REPOSITORY https://github.com/eprover/eprover

LICENSE Dual GPL-2.0-or-later and LGPL-2.1-or-later; verified.

COMPATIBILITY Use as a standalone CLI. Electing GPLv3 terms adds an express patent grant at copyleft cost; document the election and pin a release.

PATENT EVIDENCE No superposition patent was found. The lack of a default express grant and ongoing prover research keep the entry at T4.

Coq / Rocq — constructive proof assistant

T4 · MEDIUM

PAPER Bertot and Castéran, *Coq'Art*, Springer 2004.

REPOSITORY https://github.com/coq/coq

LICENSE LGPL-2.1; verified; no express patent grant.

COMPATIBILITY Link as a library sidecar and preserve replacement/modification rights.

PATENT EVIDENCE No patent was found on the Calculus of Inductive Constructions or core prover mechanisms. Modern extensions remain an active field, so foundational age does not justify T2.

Declarative engines remain practical for rules, lineage, and combinatorial configuration

SWI-Prolog — Prolog with constraints and tabling

T4 · MEDIUM

PAPER Colmerauer and Roussel, *The Birth of Prolog*; technique roots 1972–73.

REPOSITORY https://github.com/SWI-Prolog/swipl-devel

LICENSE BSD-2-Clause; verified; no express patent grant.

COMPATIBILITY Embeds directly into Apache-2.0.

PATENT EVIDENCE WAM improvement patents US 5,386,520, US 5,274,820, and US 4,546,432 are expired; WAM itself was not patented. No post-2010 engine patent, continuation, or NPE activity was found. Modern extensions keep the entry at T4.

Clingo — answer-set programming

T4 · MEDIUM

PAPER Gebser et al., *AI Communications* 24(2), 2011; rooted in Gelfond and Lifschitz 1988.

REPOSITORY https://github.com/potassco/clingo

LICENSE MIT; verified; no express patent grant.

COMPATIBILITY Embeds directly into Apache-2.0.

PATENT EVIDENCE No answer-set-programming patent was found. Conflict-driven ASP solving was publicly described by 2012.

Soufflé — compiled parallel Datalog

T1 · HIGH

PAPER Ceri, Gottlob, and Tanca, IEEE TKDE 1989.

REPOSITORY https://github.com/souffle-lang/souffle

LICENSE UPL-1.0; verified; unusually broad express patent grant.

COMPATIBILITY Use as a build-time compiler emitting C++; it can also combine directly, subject to UPL notices.

PATENT EVIDENCE Microsoft EP 3,861,459 B1 and US 11,645,285 B2 concern recursive SQL/DBMS planning and matter mainly if building a new recursive evaluator. The existing compiler can remain at arm's length.

GNU Prolog — native-code Prolog compiler

T1 · HIGH IF V3 ELECTED

PAPER No canonical paper; cite the canonical project.

REPOSITORY https://github.com/didoudiaz/gprolog

LICENSE Dual LGPL-3.0-or-later or GPL-2.0-or-later; verified.

COMPATIBILITY Elect LGPLv3 or GPLv3-or-later in writing, then keep a compile-to-executable or subprocess boundary.

PATENT EVIDENCE No modern engine-specific patent was identified. The T1 rating depends on a documented v3 election.

Explainable rules are useful; compiled Rete is the claim hotspot

Drools — production ReteOO rules

T1 · HIGH + CAUTION

PAPER Forgy, *Rete: A Fast Algorithm for the Many Pattern/Many Object Pattern Match Problem*, AI 19, 1982.

REPOSITORY https://github.com/apache/incubator-kie-drools

LICENSE Apache-2.0; verified; express §3 grant.

COMPATIBILITY Merges directly into Apache-2.0.

PATENT EVIDENCE Honeywell US 8,001,070 B2, active to April 18, 2030, names Drools as prior art and claims a compiled object-oriented Rete embodiment. Drools ReteOO predates the filing, supporting the basic technique, but ahead-of-time compilation warrants counsel review.

OpenL Tablets — spreadsheet-authored rules

T1 · MEDIUM-HIGH

PAPER No peer-reviewed paper; cite project documentation.

REPOSITORY https://github.com/openl-tablets/openl-tablets

LICENSE LGPLv3; verified; express grant through GPLv3 §11.

COMPATIBILITY Use as a separately declared library component.

PATENT EVIDENCE The same Honeywell compiled-Rete family is the principal flag. Review if deploying ahead-of-time compiled rule networks.

Easy Rules — sequential annotation rules

T4 · MEDIUM

PAPER No canonical paper; cite the project.

REPOSITORY https://github.com/j-easy/easy-rules

LICENSE MIT; verified; no express patent grant; maintenance mode since December 2020.

COMPATIBILITY Embeds directly into Apache-2.0.

PATENT EVIDENCE Sequential evaluation largely sidesteps the Rete thicket, but that is architectural risk reduction rather than clearance.

> **NPE signal.** US 9,311,141 B2 on Rete-based “survival rules by software agents,” active to November 2029, is held by Callahan Cellular LLC, an Intellectual Ventures affiliate named in 2024–2026 campaigns. Claims are application-specific; monitor the ownership and family. CLIPS is excluded because no official repository with a verifiable OSI license file was available; Rete II/III/NT are proprietary.

Graph-grounded inference is strongest when DB materialization stays isolated

Apache Jena — RDF, SPARQL, and rule inference

T4 · MEDIUM; T5 CARVE-OUT

PAPER W3C RDF 1.1 (2014), SPARQL 1.1 (2013), and OWL 2 (2009).

REPOSITORY https://github.com/apache/jena

LICENSE Apache-2.0; verified; express §3 grant.

COMPATIBILITY Combines directly, but isolate database-backed materializing inference.

PATENT EVIDENCE Oracle US 8,401,991 B2, active to October 4, 2030, directly covers database-based forward-chaining inference for RDFS/OWL. W3C's royalty-free policy is narrow: only Essential Claims of working-group participants. Morgan Stanley US 11,922,327 B2 and US 8,751,433 add a live improvement thicket. Prefer in-memory or query-time reasoning pending review.

HermiT — hypertableau OWL 2 DL reasoner

T4 · MEDIUM

PAPER Motik, Shearer, and Horrocks, JAIR 36, 2009.

REPOSITORY https://github.com/phillord/hermit-reasoner

LICENSE LGPL-3.0; verified; express grant through GPLv3 §11.

COMPATIBILITY Run as a separate reasoning service. Verify that the build excludes the bundled GPL-2 Java getopt port.

PATENT EVIDENCE No in-force tableau patent was found; the active field keeps the assessment at T4.

OWL API — ontology-programming interface

DEPENDENCY · APACHE OPTION

PAPER Project dependency used with HermiT.

REPOSITORY https://github.com/owlcs/owlapi

LICENSE Dual Apache-2.0/LGPL-3.0; verified.

COMPATIBILITY Elect Apache-2.0 and combine directly.

PATENT EVIDENCE The grant covers contributors' API implementation claims; reasoner and DB-materialization patents remain separate.

Pellet — OWL 2 DL reasoner

T4 · MEDIUM

PAPER Sirin et al., *Journal of Web Semantics* 5(2), 2007.

REPOSITORY https://github.com/stardog-union/pellet

LICENSE Dual AGPL-3.0 or commercial; verified.

COMPATIBILITY Use as a separate AGPL service, buy the commercial license, or prefer HermiT.

PATENT EVIDENCE No Pellet-specific asserted patent was found. It remains within the same active semantic-inference improvement field.

Exact scheduling and packing have excellent tools but vertical-specific claim traps

OR-Tools CP-SAT — industrial combinatorial optimization

T1 · HIGH

PAPER Perron and Furnon, 2019/2023; Stuckey, CPAIOR 2010.

REPOSITORY https://github.com/google/or-tools

LICENSE Apache-2.0; verified; express §3 grant.

COMPATIBILITY Combines directly; omit bundled commercial-solver wrappers unless separately cleared.

PATENT EVIDENCE No LCG/CP-SAT patent, litigation, or assertion against OR-Tools or its users was found. This is the cleanest branch choice.

Gecode — C++ constraint programming

T4 · MEDIUM

PAPER Schulte and Stuckey, TOPLAS 31(1), 2008.

REPOSITORY https://github.com/Gecode/gecode

LICENSE MIT; verified; no express patent grant.

COMPATIBILITY Combines directly into Apache-2.0.

PATENT EVIDENCE Propagation-engine architecture has 1970s–2008 prior art; no patent or litigation was found. It is the hackable core when OR-Tools is too rigid.

MiniZinc / libminizinc — solver-independent modeling

T1 · HIGH

PAPER Nethercote et al., CP 2007.

REPOSITORY https://github.com/MiniZinc/libminizinc

LICENSE MPL-2.0; verified; express §2.1(b) grant.

COMPATIBILITY Use unmodified as a toolchain component; MPL §3.3 permits a Larger Work under Apache-2.0.

PATENT EVIDENCE No modeling-language patent or assertion was found. The abstraction separates models from changing solver internals.

Choco-solver — Java CP library

T4 · MEDIUM

PAPER Prud'homme and Fages, JOSS 7(78), 2022.

REPOSITORY https://github.com/chocoteam/choco-solver

LICENSE BSD-3-Clause; verified; no express patent grant.

COMPATIBILITY Use as a Maven or Gradle dependency in Apache-2.0.

PATENT EVIDENCE No solver-internal patent or assertion was found. Modern lazy-clause generation keeps the field at T4.

> **Vertical carve-out.** US 11,797,951 B2, granted 2023 and active to about 2042, claims using constraint programming to obtain a machine-maintenance schedule. It targets the application layer, not solver internals. Review the family if the SaaS enters maintenance or field-service scheduling.

Plan-then-execute is auditable; top-k planning is the territory to avoid

Fast Downward — reference PDDL planner

T4 · MEDIUM

PAPER Helmert, JAIR 26, 2006.
REPOSITORY https://github.com/aibasel/downward
LICENSE GPLv3; verified; express §11 grant.
COMPATIBILITY Run as a PDDL-in/plans-out subprocess service.
PATENT EVIDENCE No SAS+ or causal-graph patent was found. IBM US 11,526,791 B2 and robotics task-planning patents US 11,602,847 B2 and US 11,498,212 B2 form an adjacent improvement thicket.

ENHSP — numeric planning

T4 · MEDIUM

PAPER Scala et al., ECAI 2016.
REPOSITORY https://github.com/hstairs/enhsp
LICENSE GPLv3-or-later; verified; express §11 grant.
COMPATIBILITY Keep as a separate service.
PATENT EVIDENCE No core PDDL 2.1/PDDL+ patent was found; the same active planning-improvement field applies.

Pyperplan — pedagogical STRIPS planner

T4 · MEDIUM

PAPER Zenodo 2020; originated in a 2010/11 Freiburg course.
REPOSITORY https://github.com/aibasel/pyperplan
LICENSE GPLv3; verified; express §11 grant.
COMPATIBILITY Use as a separate tool or as a reading reference for a clean Apache-2.0 reimplementation of prior-art algorithms.
PATENT EVIDENCE Its FF heuristic dates to Hoffmann and Nebel 2001. No Pyperplan-specific patent was found.

PDDL4J — Java PDDL 3.1 library

T4 · MEDIUM

PAPER Pellier and Fiorino, JETAI 30(1), 2017; repo-attested citation.
REPOSITORY https://github.com/pellierd/pddl4j
LICENSE LGPLv3; verified; express grant through GPLv3 §11.
COMPATIBILITY Link as a declared LGPL dependency.
PATENT EVIDENCE PDDL is an open community standard dating to 1998; no language patent was found. The cited HAL page was unavailable during source verification, so use the repo-attested citation.

> **Avoid top-k planning.** SymK is excluded because top-k/top-quality planning is the branch's hottest thicket: IBM US 9,740,978 B2, US 2021/0004741 A1 with continuation activity, US 11,727,289 B2, and US 10,783,441 B2. FF v2.3 is excluded because it has no license file.

Uncertainty tools are well-published; compiler and parallel-learning claims are live

PyMC — gradient-based MCMC

T4 · MEDIUM

PAPER Salvatier et al., PeerJ Computer Science 2:e55, 2016.

REPOSITORY https://github.com/pymc-devs/pymc

LICENSE Apache-2.0; verified; express §3 grant.

COMPATIBILITY Combines directly into Apache-2.0.

PATENT EVIDENCE Core methods date to Metropolis 1953, HMC 1987, and NUTS 2011/2014. Oracle US 10,496,929 B2, active to about 2034, claims a probabilistic-program compiler with modular IR emitting data-parallel code; it is adjacent to graph compilation, with continuation activity.

Stan — probabilistic language and compiler

T4 · MEDIUM

PAPER Carpenter et al., JSS 76(1), 2017.

REPOSITORY https://github.com/stan-dev/stan

LICENSE BSD-3-Clause; verified; no express patent grant.

COMPATIBILITY Combines directly.

PATENT EVIDENCE Stan predates Oracle's compiler filing, but the same PPL-compiler and parallel-inference thicket applies.

pgmpy — Bayesian networks and causal models

T4 · MEDIUM; T5-ADJACENT CARVE-OUT

PAPER Pearl 1988; SciPy 2015 and JMLR 2024 project papers.

REPOSITORY https://github.com/pgmpy/pgmpy

LICENSE MIT; verified; no express patent grant.

COMPATIBILITY Use as a dependency rather than rebuilding parallel structure learning.

PATENT EVIDENCE Microsoft US 8,005,770 B2, active to June 23, 2030, covers parallel Bayesian-network structure learning. pgmpy implements structure learning. Prefer classical published PC or score-based algorithms and review before parallelizing the learning engine.

Pyro — deep probabilistic programming

T4 · MEDIUM

PAPER Bingham et al., JMLR 20(28), 2019.

REPOSITORY https://github.com/pyro-ppl/pyro

LICENSE Apache-2.0; verified; express §3 grant.

COMPATIBILITY Combines directly into Apache-2.0.

PATENT EVIDENCE Variational inference and SVI have 1990s/2013 prior art. No Pyro-specific asserted patent was found, giving the cleanest posture among these four.

Core fuzzy inference is mature; automatic knowledge-base generation is not

scikit-fuzzy — Mamdani/Larsen inference

T4 · MEDIUM

PAPER Zadeh 1965; Mamdani and Assilian 1975.

REPOSITORY https://github.com/scikit-fuzzy/scikit-fuzzy

LICENSE BSD-3-Clause; verified; no express patent grant.

COMPATIBILITY Combines directly into Apache-2.0.

PATENT EVIDENCE The 1986–1995 apparatus wave is expired; US 5,710,868 expired fee-related in 2011. US 10,838,376 B2, active to March 2039, covers automatic fuzzy knowledge-base generation via artificial potential fields; US 11,221,593 is also live. Neither targets core FIS math.

FuzzyLite — object-oriented FIS runtime

T1 IMPLEMENTATION / T4 FIELD

PAPER No canonical paper; supports FCL and IEC 61131-7.

REPOSITORY https://github.com/fuzzylite/fuzzylite

LICENSE GPLv3 plus paid proprietary dual license; verified; express §11 grant under GPL.

COMPATIBILITY Run as a separate GPL service or buy the proprietary license.

PATENT EVIDENCE The contributor grant is medium-high confidence. The field remains T4 because active improvement patents surround automated model construction.

jFuzzyLogic — Java fuzzy-control runtime

T1 IMPLEMENTATION / T4 FIELD

PAPER No canonical paper; cite project documentation.

REPOSITORY https://sourceforge.net/projects/jfuzzylogic/

LICENSE Dual Apache-2.0/LGPLv3 according to SourceForge metadata; partially verified.

COMPATIBILITY Elect Apache-2.0 to combine directly; Java-only and dormant since about 2015.

PATENT EVIDENCE The Apache option would provide the branch's best license posture, but direct repository-file verification remains incomplete. Core FIS exposure matches scikit-fuzzy.

The bridge back to modern ML is useful and newly patent-active

DeepProbLog — differentiable probabilistic logic

T4 · MEDIUM

PAPER Manhaeve et al., DeepProbLog, NeurIPS 2018.

REPOSITORY https://github.com/ML-KULeuven/deepproblog

LICENSE Apache-2.0; verified; express §3 grant.

COMPATIBILITY Combines directly into Apache-2.0.

PATENT EVIDENCE No DeepProbLog-targeting patent was found. Google US 11,954,442 B2, granted 2024 and active to 2041, claims a Neural Symbolic Reader for an NLP embodiment. IBM US 2024/0054317 A1 and Bosch US 2024/0330645 A1 are pending.

LTNtorch — Logic Tensor Networks

T4 · MEDIUM

PAPER Badreddine et al., Logic Tensor Networks, AIJ 2022.

REPOSITORY https://github.com/tommasocarraro/LTNtorch

LICENSE MIT; verified; third-party reimplementation with no express patent grant.

COMPATIBILITY Combines directly into Apache-2.0.

PATENT EVIDENCE No technique-targeting patent was found. Active 2026 preprints indicate thicket growth, and the third-party license carries little contributor-patent value.

Scallop — differentiable probabilistic Datalog

T4 · MEDIUM

PAPER Li et al., Scallop, PLDI 2023; NeurIPS 2021 precursor.

REPOSITORY https://github.com/scallop-lang/scallop

LICENSE MIT; verified; no express patent grant.

COMPATIBILITY Combines directly into Apache-2.0.

PATENT EVIDENCE No Scallop-specific patent was found. As the newest entry, it has the strongest unpublished-application blind spot.

pracmln — Markov Logic Networks

T4 · MEDIUM-LOW

PAPER Richardson and Domingos, *Machine Learning* 62(1–2), 2006.

REPOSITORY https://github.com/danielnyga/pracmln

LICENSE BSD-2-Clause; verified; no express patent grant.

COMPATIBILITY Combines directly; treat the small single-author project more as a reference than a production runtime.

PATENT EVIDENCE US 8,862,523, granted 2014, claims MLN-based relational learning and creates this branch's most pronounced old-core/live-improvement pattern. Alchemy is excluded as non-OSI; NeurASP is excluded for lack of a license file.

PART III / ARCHITECTURE MAP

Twenty-four candidates combine directly; eleven belong behind a license boundary

Combine into an Apache-2.0 tree

OR-Tools; MiniZinc as an MPL component; Lean 4; Z3; MiniSAT; CaDiCaL; Drools; PyMC; Pyro; Stan; pgmpy; scikit-fuzzy; DeepProbLog; LTNtorch; Scallop; pracmln; SWI-Prolog; Clingo; Soufflé; Gecode; Choco-solver; Easy Rules; jFuzzyLogic under its Apache option; and OWL API under its Apache option.

Keep as a separate service or component

Fast Downward; ENHSP; Pyperplan; PDDL4J as an LGPL dependency; E; Coq as a sidecar; HermiT as an LGPL service; Pellet as an AGPL service or under a commercial license; GNU Prolog behind compile-to-executable boundaries; FuzzyLite as a GPL service or paid buyout; and OpenL Tablets as an LGPL component.

Positive surprises. Lean 4 combines Apache-2.0 with the cleanest overall search; Soufflé's UPL-1.0 carries the broadest express grant found; OR-Tools combines Apache-2.0 with no solver patent found; the closest CDCL-era IBM/Google patent expired February 20, 2026; and jFuzzyLogic offers an Apache-2.0 dual option, subject to direct-file verification.

Count note. Part III evaluates 36 candidates: 24 combine directly and 11 sit behind license boundaries. Apache Jena is the remaining carve-out—use in-memory or query-time reasoning directly, but isolate database materialization pending review. OWL API is counted separately from HermiT because it can be adopted independently.

The recurring design pattern

Keep mature solver, planner, and reasoner engines behind narrow interfaces; expose models, proofs, plans, or generated code rather than entangling their internals with the core. That makes license direction explicit and leaves higher-risk mechanisms replaceable.

CROSS-CUTTING FINDINGS

Pledges and networks reduce selected risks; none substitutes for product clearance

Google's 2013 pledge is not AI coverage

The verified named list covers 10 MapReduce patents and 79 IBM/CA data-center patents. It does not cover Agent Development Kit, LLM orchestration, retrieval, reinforcement learning, embeddings, or model serving. Do not cite it as an AI patent shield.

OIN protects a defined Linux System, not every workload on Linux

Open Invention Network's cross-license is tied to its Linux System package list. It does not credibly clear LLM agent orchestration, serving runtimes, RL libraries, or embedding methods. It remains useful for infrastructure-layer exposure.

LOT is real troll defense, not operating-company immunity

LOT Network reports roughly 2,250-plus members and 3.5 million patents, with free participation at revenues up to $25 million. Its protection activates when member patents transfer to patent-assertion entities. It does not neutralize operating-company claims from Baidu, Google, C3.ai, or FriendliAI.

The software-patent eligibility shield is weakening

Bloomberg Tax reported that a December 2025 USPTO memorandum integrated *Desjardins* into the MPEP and directed examiners that claims describing concrete improvements to computer functionality can clear *Alice* step 2A. Forward-looking T4 assessments deserve more caution than they did a few years ago.

Interpretation. Public prior art blocks later claims only to the extent it actually discloses them. Memberships, pledges, license grants, expiration, and invalidity doctrines each cover different failure modes; stack them rather than treating any one as universal protection.

WATCH-LIST

Four moving targets can materially change the risk map

1. Many Worlds v. OpenAI. Eastern District of Texas, filed around September 2, 2026. Five patents with March 2011 priority target retrieval, personalization, and content generation. The accused products are OpenAI's, but the assertion profile is directly relevant to RAG products.

2. NVIDIA DoRA application. Track prosecution of US 2025/0252301 A1 if the product uses weight-decomposed low-rank adaptation. It is distinct from base LoRA.

3. FriendliAI continuous batching. The family is still issuing, including US 12,524,266 B2 in 2025 and a granted European counterpart. Before SaaS launch, verify ownership, assignment, current family members, and claim scope against the actual scheduler.

4. Diffusion-model filings. Patent families are accelerating, with recent grants and applications from Google, Adobe, Baidu, Tencent, and others. Expect the landscape to look materially different in 12–24 months.

Unverified implementation watch

Recursive Synthetic Terminal Tasks: paper only; no public reference implementation found. **TensorRT-LLM:** project practice indicates Apache-2.0 headers, but the root license was not directly verified in the earlier pass. **AgentBench:** third-party evidence said Apache-2.0, but direct verification was not completed. Keep all three outside the verified shortlist.

Cadence. Review this page quarterly during active product design, then perform a full technique and claim refresh annually and immediately before launch.

CLASSICAL-AI WATCH-LIST

Seven improvement families deserve targeted monitoring

5. Semantic inference: Oracle US 8,401,991 B2, active to October 4, 2030, against database-materialized RDFS/OWL inference.

6. Rete compilation and NPE ownership: Honeywell US 8,001,070 B2 to April 2030; Callahan Cellular/IV affiliate US 9,311,141 B2 to November 2029.

7. PPL compilers: Oracle US 10,496,929 B2, active to about 2034, plus continuations, on modular probabilistic-program IR that emits data-parallel code.

8. Neuro-symbolic systems: Google US 11,954,442 B2 to about 2041, plus pending IBM and Bosch applications.

9. Maintenance scheduling: US 11,797,951 B2 to about 2042 if the SaaS applies constraint programming to maintenance or field-service schedules.

10. Top-k planning: IBM US 9,740,978 B2, US 2021/0004741 A1 continuations, US 11,727,289 B2, and US 10,783,441 B2.

11. Parallel Bayesian-network learning: Microsoft US 8,005,770 B2 to June 2030. Avoid an independent parallel structure-learning engine without review.

HOW PATENT HOLDERS PUBLICLY GRANT FREE USE / FAMILY A

Your own patents can be opened through licenses, pledges, and reciprocal networks

Mechanisms differ in legal strength, scope, beneficiaries, conditions, and transfer treatment. Read the operative text and patent schedule, not the announcement headline.

1. Open-source license patent grants

Apache-2.0 §3, GPLv3 §11, MPL-2.0 §2.1(b), EPL-2.0 §2, and UPL-1.0 automatically license covered contributor claims to every code recipient. No separate paperwork is needed: the software license is the grant. This is usually the strongest software-specific option.

2. Unilateral non-assertion pledges

Examples include Tesla's 2014 good-faith-use pledge, IBM's 2005 pledge covering 500 patents, Red Hat's Patent Promise, and Google's 2013 pledge. A bare promise may rely on equitable estoppel rather than contract and can be softer than a license. Google's verified list contains only 10 MapReduce and 79 IBM/CA data-center patents — no AI coverage.

3. Field- and time-limited pledges

The 2020 Open COVID Pledge showed how a holder can limit non-assertion by field and date — for example, uses to diagnose, prevent, or treat COVID-19 until a stated endpoint.

4. Defensive Patent License

The EFF-backed DPL is "copyleft for patents": members grant each other royalty-free licenses, and a member that sues another loses its protection. It is the closest model here to a reciprocal patent commons with teeth.

Networks and defensive publication cover different failure modes

5. Standards commitments

W3C is royalty-free by default for Essential Claims of participants. IEEE and ITU commonly use FRAND/RAND: not necessarily free, but enforceable public constraints on monetization.

6. Open Invention Network

OIN provides a royalty-free cross-license among 4,000-plus members for the defined Linux System. Microsoft joined in 2018 with a reported 60,000-patent contribution. It helps at the infrastructure layer, not outside the package definition.

7. LOT Network

A license-on-transfer protects members when a member patent reaches an assertion entity. It is troll defense only, not operating-company immunity, and is free below $25 million revenue.

8. Defensive publication

An enabling paper, arXiv preprint, public repository, detailed blog post, IP.com disclosure, or Research Disclosure record can become prior art and destroy novelty worldwide. It blocks everyone — including the publisher — from patenting the disclosed technique later.

Publication hygiene

Describe mechanisms, alternatives, inputs, outputs, and edge cases; publish early; keep immutable dates, authorship, repository history, and archived copies; use a formal service for valuable disclosures that do not warrant a paper.

Eric's belt-and-suspenders pattern. Publish openly and early so later applicants face prior art, then license all code under Apache-2.0 so §3 automatically grants Eric's contributor claims to every recipient.

Present versus future. Grants and publications shape the future landscape. They do not replace freedom-to-operate review against patents already in force. Existing licenses and pledges, expiry, invalidity, and claim-specific counsel address the present.

General information, not legal advice. Draft and evaluate grants, pledges, commitments, and publication timing for the relevant jurisdiction and business facts.

DEFENSIVE POSTURE

Publish early, join the right networks, and buy narrow legal answers

1. Create prior art deliberately

Every sufficiently detailed paper, engineering post, and public commit can block future patents on the disclosed technique. For ideas not worth a full paper, consider IP.com or another defensive-publication aggregator.

2. Join LOT at incorporation

At the stated revenue level, participation is free. It is worthwhile troll-defense hygiene, but only for member patents later transferred to assertion entities.

3. Consider OIN membership

The royalty-free cross-license can reduce exposure in the Linux infrastructure layer supporting the SaaS stack. Do not extrapolate that coverage to AI techniques.

4. Keep openness in perspective

Always-public source makes GPL and AGPL obligations manageable, including a compliant network source link. It does not answer third-party patents; licenses, pledges, expiry, and counsel do.

5. Reduce exposure structurally

Prefer PPO, SAC, or TD3 over DQN. Avoid A3C/IMPALA where practical or use a Google Apache-2.0 implementation if available. Track DoRA. Keep GPL/AGPL components separately licensed so the core can remain Apache-2.0.

6. Preserve an audit trail

Record papers, claims reviewed, license texts, notices, dependencies, design alternatives, and the exact release adopted. Revisit the record when mechanisms change.

Highest-value counsel engagements, in order

1. Baidu US 10,332,509 against any end-to-end speech recognizer.
2. FriendliAI's continuous-batching family against the inference scheduler.
3. C3.ai US 12,111,859 against multi-agent orchestration.
4. Google US 9,037,464 if shipping gensim-style embeddings — or use Google's Apache-2.0 word2vec implementation instead.

Practical position. The safest path is not "avoid all patents." It is to combine contributor grants, expired foundations, early publication, careful algorithm selection, and narrow counsel review at the few points where the evidence identifies real exposure.

COVERAGE NOTES

The list is intentionally selective where credible paper-backed implementations are thin

MPL-2.0

Thin but credible: Coqui STT and TTS are the two notable paper-backed entries. File-level copyleft permits direct combination when covered files retain MPL.

EPL-2.0

The search found hobby projects but no implementation with both a real paper and meaningful adoption. The document does not pad this family with weak candidates.

LGPL

Only gensim qualified, under LGPL-2.1, which lacks an express patent grant. No notable paper-backed LGPL-3.0 ML implementation was found.

Excluded terms

Non-commercial, source-available, and Server Side Public License projects are excluded. This reference focuses on licenses compatible with commercial SaaS and an always-public codebase.

Release checklist. Re-check the exact revision, file headers, NOTICE obligations, bundled dependencies, model and dataset terms, active patent families, and relevant jurisdictions. Then obtain a product-specific freedom-to-operate review before commercialization.

How to use the document

Start with T5 and the watch-list when choosing architecture. Prefer T1+T2 anchors where they fit. Use T1-backed T4 entries as candidates, not certifications. Treat MIT entries as technically useful but patent-unshielded. Revisit any conclusion when the product changes how a technique is implemented.

Edition assumptions. The jurisdictional default is the United States; the intended product model is commercial SaaS with publicly available source; the preferred core license is Apache-2.0 or MIT. Different facts can change both license compatibility and patent exposure.